

RAMAGUNDAM, India

WMO: 430860

Lat: 18.767N

Lon: 79.433E

Elev: 156

StdP: 99.46

Time Zone: 5.50 (IND)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	13.3	14.4	4.0	5.1	27.9	6.5	6.1	27.1	3.1	26.7	2.5	27.0	0.0	230	0.321

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	11.4	42.8	25.3	41.6	25.0	40.3	25.1	28.7	36.5	28.0	35.3	27.5	34.1	1.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	23.1	31.7	26.4	22.3	31.0	25.9	21.7	30.4	94.4	36.8	91.1	35.6	88.4	33.8	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
3.7	3.2	2.6	10.8	44.6	1.1	1.3	10.0	45.6	9.4	46.4	8.8	47.1	8.0	48.0		
			8.9	29.5	1.2	0.5	8.0	29.9	7.3	30.2	6.6	30.4	5.7	30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	23.0	26.3	29.6	33.3	35.4	32.3	28.8	28.1	28.5	27.7	24.9	22.8
	DBStd	4.26	1.95	2.01	2.05	1.86	2.21	3.28	1.93	1.56	1.26	1.49	1.83	1.92
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6714	403	456	607	699	788	669	583	562	556	548	447	396
	CDD18.3	3673	145	223	348	449	529	419	324	304	306	290	197	137
	CDH23.3	46147	1172	2228	4451	6909	8745	6282	3851	3375	3472	2948	1644	1069
	CDH26.7	24988	373	1075	2506	4568	6279	3949	1623	1193	1318	1206	590	306
Wind	WSAvg	0.7	0.6	0.9	0.9	1.2	1.1	1.1	0.8	0.7	0.5	0.4	0.4	0.3
Precipitation	PrecAvg	972	13	11	13	15	15	147	255	246	167	75	13	4
	PrecMax	1469	73	69	121	92	114	277	462	612	401	244	111	38
	PrecMin	593	0	1	0	0	0	24	65	66	7	0	0	0
	PrecStd	241	23	16	28	24	26	73	119	135	106	65	28	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.7	36.5	39.5	42.8	44.8	43.6	35.6	34.4	35.2	34.9	32.9	31.8
		MCWB	20.2	20.8	22.5	24.6	25.1	26.1	25.5	26.0	25.9	24.3	22.6	20.5
	2%	DB	31.1	34.9	38.2	41.5	43.5	41.6	34.5	33.1	33.6	33.7	31.8	30.5
		MCWB	19.9	21.1	23.0	24.1	25.2	25.8	25.4	25.8	25.8	24.6	22.1	20.1
	5%	DB	30.0	33.7	37.0	40.4	42.3	39.8	33.5	32.2	32.6	32.8	30.9	29.5
		MCWB	19.5	21.0	22.9	24.4	25.1	25.6	25.4	25.7	25.6	24.2	21.6	19.6
	10%	DB	28.9	32.4	35.7	39.2	41.2	38.0	32.4	31.3	31.7	31.9	30.0	28.5
		MCWB	19.1	20.7	22.5	24.4	25.3	25.4	25.5	25.6	25.6	23.8	21.2	19.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.5	25.2	27.3	29.1	29.5	29.1	28.0	27.7	27.7	27.6	25.7	23.7
		MCDB	28.2	30.3	33.2	36.4	36.5	36.7	32.0	31.3	31.6	32.1	29.9	28.1
	2%	WB	22.5	23.6	26.1	28.0	28.9	28.2	27.2	27.2	27.3	26.7	24.9	22.5
		MCDB	27.2	30.5	32.9	35.5	37.7	35.8	31.3	30.7	31.2	31.0	29.3	27.0
	5%	WB	21.6	22.7	25.2	27.1	28.2	27.6	26.7	26.7	26.8	26.1	24.2	21.7
		MCDB	26.8	29.7	32.2	35.2	37.2	34.7	30.8	30.2	30.7	30.3	28.3	26.5
	10%	WB	20.8	22.1	24.6	26.4	27.5	27.0	26.3	26.3	26.4	25.6	23.4	20.9
		MCDB	26.5	29.0	31.8	34.8	36.5	33.7	30.3	29.8	30.1	29.6	27.3	25.9
Mean Daily Temperature Range	5% DB	MDBR	12.9	12.7	12.6	12.1	11.4	8.1	5.7	5.2	6.2	8.9	11.4	12.9
		MCDBR	13.9	14.3	14.2	13.3	12.4	10.4	7.4	6.7	7.8	10.3	11.9	13.7
		MCWBR	4.8	4.2	4.5	4.6	4.4	3.4	2.6	2.7	2.9	3.6	3.9	4.8
	5% WB	MCDBR	10.9	11.9	11.6	11.5	11.1	9.0	6.6	6.0	6.8	8.1	8.7	10.6
		MCWBR	4.8	4.5	4.7	4.9	4.7	3.6	2.9	2.8	3.1	3.5	3.7	4.5
Clear-Sky Solar Irradiance	taub	0.481	0.499	0.566	0.593	0.623	0.637	0.589	0.558	0.557	0.540	0.492	0.463	
	taud	2.018	1.987	1.845	1.834	1.775	1.780	1.898	1.991	1.987	1.997	2.064	2.085	
	Ebn at Noon	791	803	764	746	716	699	735	763	760	757	775	793	
	Edh at Noon	164	178	212	216	226	223	198	182	181	174	155	149	
All-Sky Solar Radiation	RadAvg	4.71	5.56	6.18	6.74	6.74	5.09	4.07	4.08	4.85	5.10	4.80	4.51	
	RadStd	0.21	0.20	0.33	0.22	0.20	0.47	0.52	0.41	0.34	0.28	0.29	0.25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	+0.25	N/A	-1.67	+0.96	N/A	N/A	N/A	N/A	+140	+139

Nomenclature: See separate page